

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2023

Course Code: MAT101R01

Course: **ENGINEERING MATHEMATICS - I**

QP No. :UA184-A

Duration: 3 hours

Max. Marks: 100

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. If $\frac{\sin \theta}{\theta} = \frac{19493}{19494}$, prove that θ is equal to 1° nearly.
2. Separate $\cos(x + iy)$ into their real and imaginary parts.
3. Solve the equation $x^4 + 2x^3 - 16x^2 - 22x + 7 = 0$, given that $2 + \sqrt{3}$.
4. Find the sum of series $\log a - \frac{1}{2!}(\log a)^2 + \frac{1}{3!}(\log a)^3 + \dots$
5. If the sum of two eigen values and trace of a 3×3 matrix A are equal, Find the value of $|A|$.
6. Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$
7. State Vieta's formula for the polynomial equation of degree 3.
8. Evaluate $\int_0^a \int_0^{\sqrt{a^2 - x^2}} dy dx$.

9. Find the area between the circle $x^2 + y^2 = a^2$ and the line $x + y = a$ lying in the first quadrant by double integration.
10. Change the order of integration $\int_0^a \int_0^x f(x, y) dy dx$.

PART- B

Answer all the questions

4 x 15 = 60 Marks

11. a) Prove that $\frac{\sin 7\theta}{\sin \theta} = 7 - 56\sin^2 \theta + 112\sin^4 \theta - 64\sin^6 \theta$. (8)
- b) Solve the equation $\cos\left(\frac{\pi}{6} - x\right) = 0.88$. (7)

(OR)

12. a) Expand $\sin^3 \theta \cos^5 \theta$ in a series of sines of multiples of θ . (8)
- b) If $x + iy = \cos(u + iv)$, where x, y, u, v are real, prove that $(1 + x)^2 + y^2 = (\cosh v + \cos u)^2$. (7)
13. a) Reduce the quadratic form $x_1^2 + 2x_2x_3$ into a canonical form by means of an orthogonal transformation. Also determine its nature. (8)

- b) Find the rank of the matrix $A = \begin{bmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{bmatrix}$ (7)

(OR)

14. a) Find a singular value decomposition for $A = \begin{bmatrix} 1 & -2 & 3 \\ 3 & 2 & -1 \end{bmatrix}$ (10)
- b) State Rouche's theorem and discuss about consistency and inconsistency of a system of equations. (5)

15. a) Solve the equation $x^4 - 2x^3 - 23x^2 - 12x + 36 = 0$, given that the product of two of its roots is equal to the product of the other two. (8)

- b) Find the condition that the root of the equation $x^3 + px^2 + qx + r = 0$ in geometric progression. (7)

(OR)

16. a) Diminish the roots of the equation $x^4 - 4x^3 - 7x^2 + 22x + 24 = 0$ by 1 and hence solve it. (8)

- b) Solve the equation $8x^5 - 22x^4 - 55x^3 + 55x^2 + 22x - 8 = 0$. (7)

17. a) Find the area that lies inside the cardioid $r = a(1 + \cos\theta)$ and outside the circle $r = a$, by double integration. (8)

- b) Prove the relation between Beta and Gamma functions $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. (7)

(OR)

18. a) Express the volume of the sphere $x^2 + y^2 + z^2 = a^2$ as a volume integral and hence evaluate it. (8)

- b) Change the order of integration $\int_0^1 \int_y^{2-y} xy dy dx$ and evaluate it. (7)

PART-C

Answer the following

1 x 20 = 20 Marks

19. a) Determine the nature of roots $x^7 + 8x^5 - x + 9 = 0$. (10)
- b) Evaluate $\iiint_V xyz \, dx dy dz$, where V is the region of space inside the tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (10)